



Game Design Document: Neural Matrix

1. Game Overview

Title: Neural Matrix **Genre:** Hyper-casual, Brain Training, Puzzle **Platform:** Web (Mobile-first, Desktop compatible) **Engine:** HTML5, CSS3, JavaScript (Three.js for 3D iterations) **Core Concept:** A high-intensity cognitive training game where players must memorize and recreate patterns of active nodes within a dynamic grid.

2. Target Audience

- **Hyper-casual Gamers:** Looking for short, engaging 1-3 minute loops.
- **Self-Improvement/Brain Training Users:** Individuals looking to improve spatial and sequential working memory.
- **Sci-Fi/Cyberpunk Fans:** Players drawn to neon aesthetics, synth audio, and "hacking" motifs.

3. Core Gameplay Loop

1. **Observation Phase:** A 3D matrix appears. A sequence of nodes illuminates briefly, accompanied by synthesized audio cues.
2. **Execution Phase:** A stress timer begins decaying. The player must select the correct nodes before time runs out.
3. **Resolution Phase:**
 - *Success:* Combo multiplier increases, time is added, and the player advances to a harder level (larger grid or more nodes).
 - *Failure/Timeout:* The sequence is failed, the combo drops, and the game ends, bringing up the local leaderboard.

4. Key Mechanics

- **Spatial Mode (Classic):** Players must remember *which* nodes lit up, but the order of selection does not matter.
- **Sequential Mode (Hardcore):** Players must remember the exact *order* in which the nodes illuminated and select them accordingly.
- **Momentum & Combos:** Rapidly executing the correct sequence within a 1-second window per click builds a combo multiplier, drastically increasing point yields.
- **The Stress Timer:** A decaying bar that scales based on difficulty. Correct clicks add a small time bonus, forcing players to balance speed with accuracy.
- **Dynamic Difficulty Scaling:** Every 2 levels, the sequence gets longer. Every 3 levels, the grid expands (3x3 -> 4x4 -> 5x5).

5. Aesthetics & Audio

- **Visual Style:** Wireframe edges, dark voids, and emissive neon lighting powered by WebGL (Three.js).
- **Theme Engine:** Players can toggle between color palettes (e.g., Cyberpunk, Synthwave, Monochrome) to suit visual preferences or colorblindness needs.
- **Procedural Audio:** Using the Web Audio API to generate dynamic sine, square, and sawtooth waves for hovers, selections, errors, and level-ups, eliminating the need for external MP3 assets.

6. Future Expansion (Monetization & Metagame)

- **Currency:** Players earn "Data Fragments" based on their score.
- **Storefront:** Spend fragments to unlock new node shapes (spheres, pyramids), custom synth-kits (lo-fi, heavy bass), or background environments.
- **Backend Database:** Implement Firebase/Supabase for global cross-platform leaderboards and "Daily Uplink" seeded challenges.